

Youtube Artificial Intelligence Censorship

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Abstract— Youtube is one of the world’s biggest video sharing platforms, and although Youtube gives its users a large amount of freedom within what they are able to post on the website, Youtube still needs to have strong censorship guidelines within its community. Youtube utilizes a combination of artificial intelligence systems and human reviews to be able to enforce its censorship guidelines. This paper will discuss the growing issue of Youtube’s artificial intelligence over censoring people within the community and how MapReduce reuse can be implemented within Youtube’s current algorithm to improve the issue.

1 PROBLEM SPACE

1.1 Problem and Context

Almost everyone in society is familiar with Youtube and has utilized the video sharing platform at least one or twice in their lives. The only reason that it has become a stable activity in so many lives is due to the amount of content that is available to be consumed on the site. There are millions and millions of users who post videos daily. However, with so many videos posted every single day, Youtube needs to employ the use of both human reviewers and automated artificial intelligence systems to be able to maintain good community guidelines between its users. Youtube’s censorship policies prohibits users posting videos with copyrights, pornography, racism, illegal acts, hate speech, and many more. With such a multitude of videos to go through at all times, it would be physically impossible to have enough human reviewers to scour every video and identify every issue. This has led Youtube to heavily depend on automated systems to be able to catch a majority of these violations, so that videos with these issues can be removed. The issue that has arisen within the last few years is the over censorship that Youtube has seen since its AI systems have been automated to be able to remove, ban, and censor users and videos completely on its own. Around 11 million videos were removed from the site between April and June in 2020 when this issue began. The number of videos that were appealed and reinstated were close to 200 thousand (Vincent, 2020). Since then, these numbers have not gone down much over the years. Within the last quarter, there have been over 8 million videos removed from Youtube, with over 98% of those videos being automated system removals and a total of 50 thousand reinstatements so far. This can be seen from the graph shown below.

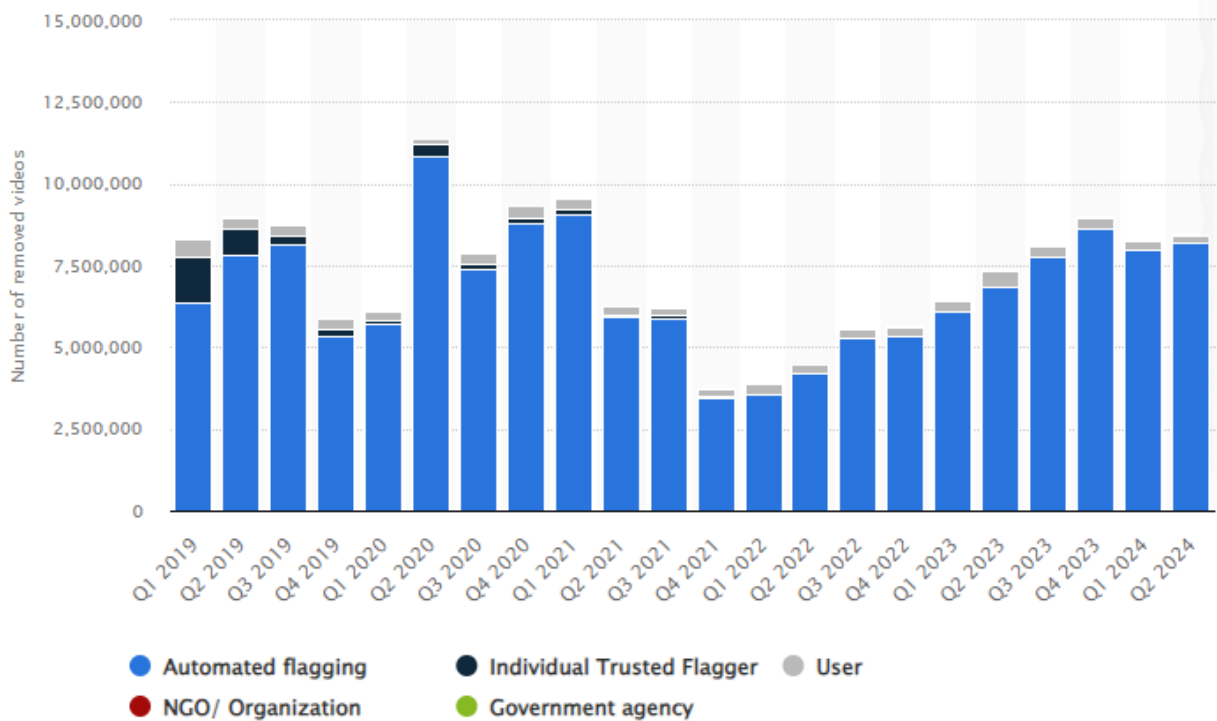


Figure 1—Graph of Number of Removed Videos from Youtube. [Source](#)

As you can see the ratio of the number of videos removed and the ratio of the number of videos reinstated from Youtube from automated systems has stayed relatively consistent for the past few years. This gives a clear and present issue that Youtube’s censorship team’s use of automated censorship systems is still over censoring videos within the community. Youtube’s current algorithm utilizes what is known as hash matching to be able to identify videos with censorship violations within it. The process involves generating unique digital fingerprints, also known as hashes for known videos, including flagged or copyrighted material. When a new video is uploaded, YouTube generates a hash for the video or different segments throughout the video and compares these hashes against its database of known content. If a match is found, the system flags the video for review, automatically applying censorship or copyright claims as needed (Fanello et al., n.d.). The issue with this process is that this process is not quite as effective in content moderation censorship since it is not effective with any type of modifications within the videos. Any type of minor changes will cause an entirely new hash value to be generated. This oversight is resolved with machine learning models to perform content analysis. However, ultimately this causes a massive over censoring of videos in the Youtube community instead.

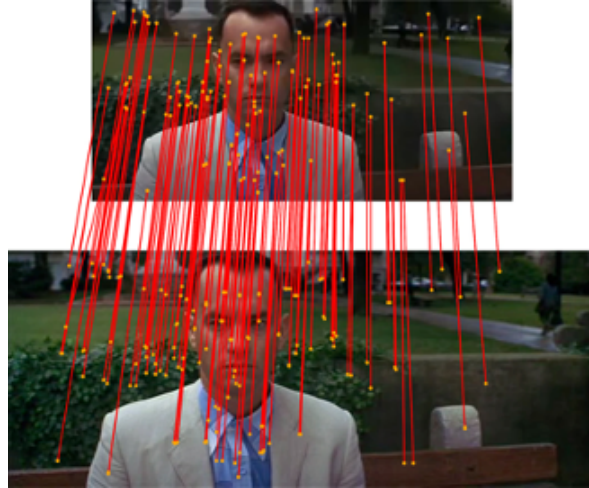


Figure 2—Image of Hash Matching. [Source](#)

1.2 Comparison

My proposed solution is to include the utilization of MapReduce within Youtube’s current censorship automated systems to be able to improve copyright detection and other violation detections. Youtube’s current system main problem is the over censorship of videos. However, by including MapReduce, there can be a secondary system that double checks the results that the hash matching system arrives at. Since Youtube wants to avoid utilizing human reviewers, the only solution is to improve the overall accuracy of Youtube’s censorship system, and MapReduce is able to resolve this issue. With MapReduce, the system will be able to detect copyright infringement within videos, detect objects through video analysis for any violations, and do image recognition through the video as well. The baseline design will be the implementation of the MapReduce system instead of Youtube’s hash matching. For further refining, an implementation of a combination of both systems can be done to further improve the results. With both systems together, the accuracy in which Youtube’s automated censorship should be able to improve, and negate the over censorship that the current system is creating.

2 BASELINE DESIGN

2.1 Scenarios

The baseline design will be tested with the following three scenarios used for validation of the MapReduce system inclusion into Youtube’s current censorship guideline system.

1. The baseline design is utilized to analyze videos that are determined to have no violations within Youtube’s community guidelines and determine if there are any violations.
 - a. This scenario is designed to be the easiest since the system will not have any violations it needs to identify.

2. The baseline design is utilized to analyze videos that are determined to have severe violations within Youtube's community guidelines and determine if there are any violations.
 - a. This scenario is designed to be average since the system will have clear and concise indications of guideline violations.
3. The baseline design is utilized to analyze a massive amount of videos that are determined to have a mixed variety of degree of violations within Youtube's community guidelines to have no violations and determine if there are any violations.
 - a. This scenario is designed to be the hardest since the system will have the most mixed variety of videos that needs to be analyzed.

2.2 Baseline Method Workflow

1. Video Uploaded to Youtube
2. Map Phase
 - a. Video is split into chunks for parallel processing
3. Generate Hash for Each Chunk
 - a. A hash is generated for each video chunk
4. Reduce Phase
 - a. The hashes of the chunks are compared against the database of known violations
5. If Match is Found
 - a. If a match is found then video is flagged and then video will be removed and the user is notified and penalized.
 - b. If no match is found then the process continues to next video

These workflow steps can be used to support all three example scenarios and will be used as the basis of how the process of the baseline design should be tested.

2.3 Architecture

Pseudo code for MapReduce Functionality of Youtube Video Analysis

function map(videoChunk):

videoHash = generateHash(videoChunk)

emit(videoHash, videoChunk)

function reduce(videoHash, videoChunks):

censoredHashDatabase = loadCensoredHashes()

if videoHash in censoredHashDatabase:

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        flagForCensorship(videoChunks)
    else:
        allow(videoChunks)

function processVideoForCensorship(videoFile):
    videoChunks = splitVideoIntoChunks(videoFile)

    for each chunk in videoChunks:
        map(chunk)

    hashBuckets = shuffleAndSort(videoChunksByHash)

    for each (videoHash, chunks) in hashBuckets:
        reduce(videoHash, chunks)

function processVideos(videoFiles):
    for each videoFile in videoFiles:
        processVideoForCensorship(videoFile)

```

2.4 Alternative Implementation and Execution

Although the baseline design should be able to improve the accuracy of Youtube's censorship systems well, there are still possible improvements that can be done to the baseline design as well. The current baseline design replaces the hash matching within Youtube's current system with MapReduce. However, with the removal of Youtube's AI and machine learning systems for the content analysis, the censorship system will be censoring a majority of the videos. To counteract this issue, refining the baseline design to incorporate both Youtube's current system and the new MapReduce system will allow for the most effective censoring of videos.

3 DESIGN REFINEMENTS

3.1 Refined Design

The refined design follows would be a combination of Youtube's current censorship system and the MapReduce implementation in the baseline design. This design follows a similar flowchart to the baseline design; however, after the MapReduce comparison of the generated hash is complete. Despite the results, the video will then undergo content analysis from Youtube's current artificial intelligence and machine learning models for any guideline violations. The results will then determine the next steps for the video. If both steps result in violations then the

video will be determined to be in violation and removed from the site. However, if only one of the systems identifies a violation in the video then this would require human review. The video then is sent towards a designated review team to look over the video. If no violations were found then the system will be able to move onto the next video.

3.2 Refined Design Workflow

1. Video Uploaded to Youtube
2. Map Phase
 - a. Video is split into chunks for parallel processing
3. Generate Hash for Each Chunk
 - a. A hash is generated for each video chunk
4. Reduce Phase
 - a. The hashes of the chunks are compared against the database of known violations
5. If Match is Found
 - a. If a match is found then video is flagged for violation and moved towards Youtube's current violation check
 - b. If no match is found video is still moved towards Youtube's current violation check
6. Youtube's Current Violation Check
 - a. AI Content Review Systems using machine learning models for an additional check for content violation. This is Youtube's current system.
7. If Violation is Found
 - a. If violation is found and match is also flagged then video will be removed and user is notified and penalized.
 - b. If violation is found and the match is not flagged then human review will be requested from the team.
 - c. If violation is not found, but is flagged then human review will be requested from the team.
 - d. If violation is not found then the process continues to next video

3.4 Comparison/Evaluation Metrics

Comparing the baseline design and the refined design, there is a clear distinction in which design is overall better. The baseline design improves upon the hash matching of Youtube's current censorship system. However, the current system also incorporates in the machine learning models to identify content violations in addition to the hash matching. The baseline design only improves upon one aspect of the current design while also not including the missing part of the current design. This will clearly leave a gap in the analysis of the videos and severely lowers the

accuracy of violation detection within the baseline system. The refined design, on the other hand, takes the best aspects of both designs and combines them together to become the most efficient version since the MapReduce and machine learning models are able to work together to identify violations. Ultimately, the evaluation of the refined design will be based on the accuracy of the violation detection compared to the baseline design and original system. For the refined design to be considered successful, the system must have an overall higher accuracy of guideline violation compared to other designs, while also keeping the over censor of videos to a minimum as well.

4 ADDITIONAL DESIGN CONSIDERATIONS

4.1 Baseline Evaluation Metrics

Like the refined design, the evaluation of the baseline design is determined through the accuracy of the identification of videos that have guideline violations. There are two metrics that the design will be analyzed and compared with the original and the refined design, which is the number of accurate videos correctly identified as having violations and the number of videos that were falsely identified as a violation. The baseline design can be considered successful if the accuracy of the correctly identified videos are higher than the original system and the number of incorrectly identified videos are lower compared to Youtube's current quarter number on reinstated videos.

4.2 Additional Considerations

Although the refined design should be an improvement to the current system, other issues that will need to be addressed to improve the system overall would be the speed and scalability of the system. Although the MapReduce already improves the speed of the system, the issue extends towards the comparison between the hash of the video and the database. As more and more videos are created, the extent of the violations held within the database will become bigger and bigger. Even though the violations are held in hashes, this issue is an ever growing database that will continue to grow large without stopping. The database needs to be improved to handle the massive amounts of data.

5 REFERENCES

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